

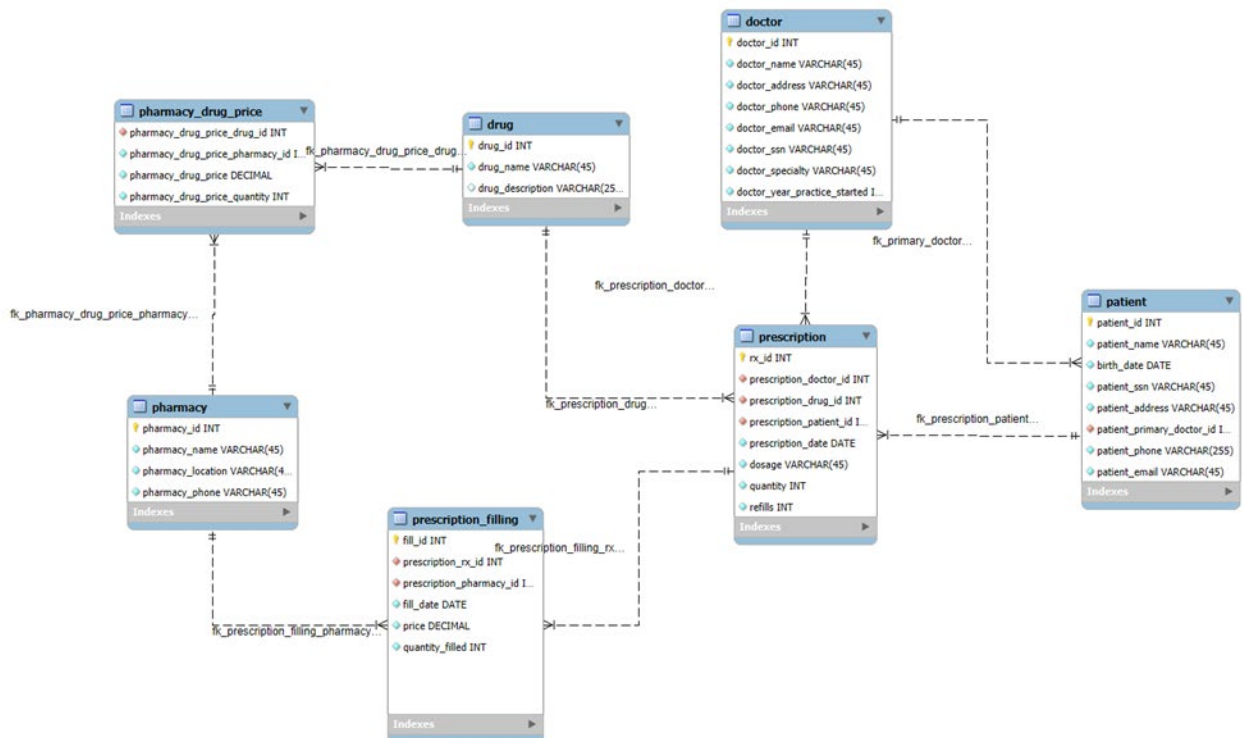
Tasneem Khokha
Saria Kabbour
Gerek Jordan

CST 363 Database Management
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Lab 19 Web App SQL (Pharmacy)

This project is a design of a database system for managing patient prescriptions in a pharmacy on a web application. The system tracks doctors, patients, drugs, pharmacies, and prescriptions, and ensures that each prescription is properly linked to the patient, doctor, and the drug being prescribed. It also allows pharmacies to manage prescription filling and provides a solution for handling medical prescriptions and refills across multiple entities. It allows doctors to write a prescription, patients to request a prescription to be filled, pharmacies to register new patients and doctors, and change patient and doctor information. Below is an ERD of a system that enforces key relationships like doctor-patient assignments, prescription tracking, and drug availability at pharmacies.

Entity-Relationship Diagram



SQL Schema

Initial Creation

```
drop database if exists prescription;  
create database prescription;  
USE prescription;
```

Creation of doctor table:

```
create table doctor(  
  id int primary key auto_increment,  
  ssn varchar(9) not null unique,  
  last_name varchar(30) not null,  
  first_name varchar(30) not null,  
  specialty varchar(30),  
  practice_since int );
```

Creation of patient table:

```
CREATE TABLE patient (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  first_name VARCHAR(30) NOT NULL,  
  last_name VARCHAR(30) NOT NULL,  
  birthdate DATE,  
  ssn VARCHAR(9) NOT NULL UNIQUE,  
  street VARCHAR(50),  
  city VARCHAR(30),  
  state VARCHAR(2),  
  zipcode VARCHAR(10),  
  doctor_id INT,  
  FOREIGN KEY (doctor_id) REFERENCES doctor(id) ON DELETE SET NULL  
);
```

Creation of drug table:

```
CREATE TABLE drug (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(50) NOT NULL UNIQUE  
);
```

Creation of prescription table:

```
CREATE TABLE prescription (  
  rxid INT PRIMARY KEY AUTO_INCREMENT,  
  patient_id INT,  
  doctor_id INT,
```

```
drug_id INT,  
quantity INT,  
date_created DATE,  
refills_remaining INT,  
FOREIGN KEY (patient_id) REFERENCES patient(id) ON DELETE CASCADE,  
FOREIGN KEY (doctor_id) REFERENCES doctor(id) ON DELETE CASCADE,  
FOREIGN KEY (drug_id) REFERENCES drug(id)  
);
```

Creation of prescription_fill table:

```
CREATE TABLE prescription_fill (  
fill_id INT PRIMARY KEY AUTO_INCREMENT,  
rxid INT,  
pharmacy_id INT,  
date_filled DATE,  
cost DECIMAL(10, 2),  
FOREIGN KEY (rxid) REFERENCES prescription(rxid) ON DELETE CASCADE,  
FOREIGN KEY (pharmacy_id) REFERENCES pharmacy(id)  
);
```

Creation of drug_cost table:

```
CREATE TABLE drug_cost (  
id INT PRIMARY KEY AUTO_INCREMENT,  
drug_id INT NOT NULL,  
pharmacy_id INT NOT NULL,  
cost DECIMAL(10, 2) NOT NULL,  
FOREIGN KEY (drug_id) REFERENCES drug(id) ON DELETE CASCADE,  
FOREIGN KEY (pharmacy_id) REFERENCES pharmacy(id) ON DELETE CASCADE  
);
```

Creation of test drugs:

```
INSERT INTO drug (name) VALUES ('Loratadine');  
INSERT INTO drug (name) VALUES ('Acetaminophen');  
INSERT INTO drug (name) VALUES ('Lisinopril');  
INSERT INTO drug (name) VALUES ('Lovastatin');
```

Creation of test pharmacies:

```
INSERT INTO pharmacy (name, address, phone) VALUES ('CVS', '111 Main St, Seaside, CA',  
'123-4567');
```

```
INSERT INTO pharmacy (name, address, phone) VALUES ('Rite Aid', '222 Ocean Ave, Seaside, CA', '765-4321');
```

-CVS has ID of 1.

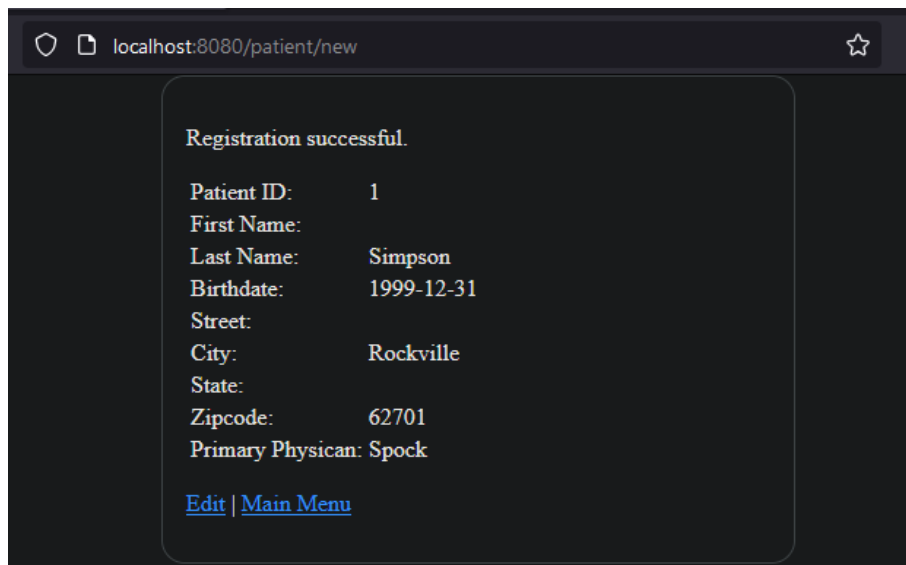
-Rite Aid has ID of 2.

Creation of test drugs:

```
INSERT INTO drug_cost (drug_id, pharmacy_id, cost) VALUES (1, 1, 10.99);  
INSERT INTO drug_cost (drug_id, pharmacy_id, cost) VALUES (2, 1, 9.99);  
INSERT INTO drug_cost (drug_id, pharmacy_id, cost) VALUES (3, 2, 15.99);  
INSERT INTO drug_cost (drug_id, pharmacy_id, cost) VALUES (4, 2, 19.99);
```

Screenshots

1. Registration of a new patient with last name “Simpson”, city “Rockville”, zip code 62701 and a doctor with name “Spock”.



97 • `SELECT * FROM patient;`

	id	first_name	last_name	birthdate	ssn	street	city	state	zipcode	doctor_id
▶	1		Simpson	1999-12-31	000112222		Rockville		62701	1

2. Registration attempt for a new patient with last name “Burns” but with a doctor name that does not exist. Screen of the patient register form with the error message:

localhost:8080/patient/new

Register as new user

Doctor not found.

Your SSN: 000112222

Your First Name: Jack

Your Last Name: Burns

Birth Date: 01/01/2000

Street: Main

City: Los Angeles

State: California

Zipcode: 00000

Primary Physician
Name: Strange

Register

3. Creation of a prescription for the patient “Simpson” and doctor “Spock” for a drug “lisinopril” and quantity 90. Show the screen with the success message and prescription display:

localhost:8080/prescription

Prescription created successfully.

Rx: 2

Doctor ID: 1

First Name: Spock

Last Name: Spock

Patient ID: 1

First Name: Jack

Last Name: Simpson

Drug: Lisinopril

Quantity: 90

Refills remaining: 0

Pharmacy ID: 0

Name:

Address:

Phone:

Date Filled:

Cost: \$

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4. Attempt to create a prescription with an invalid drug name. Screen with the create prescription form and error message:

localhost:8080/prescription

New Prescription Form

Error: Invalid drug name.

Doctor ID:

Doctor First Name:

Doctor Last Name:

Patient ID:

Patient First Name:

Patient Last Name:

Drug Name:

Quantity:

Number of refills:

5. Attempt to fill a prescription with an invalid pharmacy name

localhost:8080/prescription/fill

Request Prescription to be filled.

Error: Invalid pharmacy name.

Rx:

Patient Last Name:

Pharmacy Name:

Pharmacy Address:

6. Attempt to fill a prescription with an invalid rxid.

localhost:8080/prescription/fill

Request Prescription to be filled.

Error: Invalid RxID.

Rx: 100

Patient Last Name: Simpson

Pharmacy Name: CVS

Pharmacy Address: 111 Main Street

Request Fill for Prescription

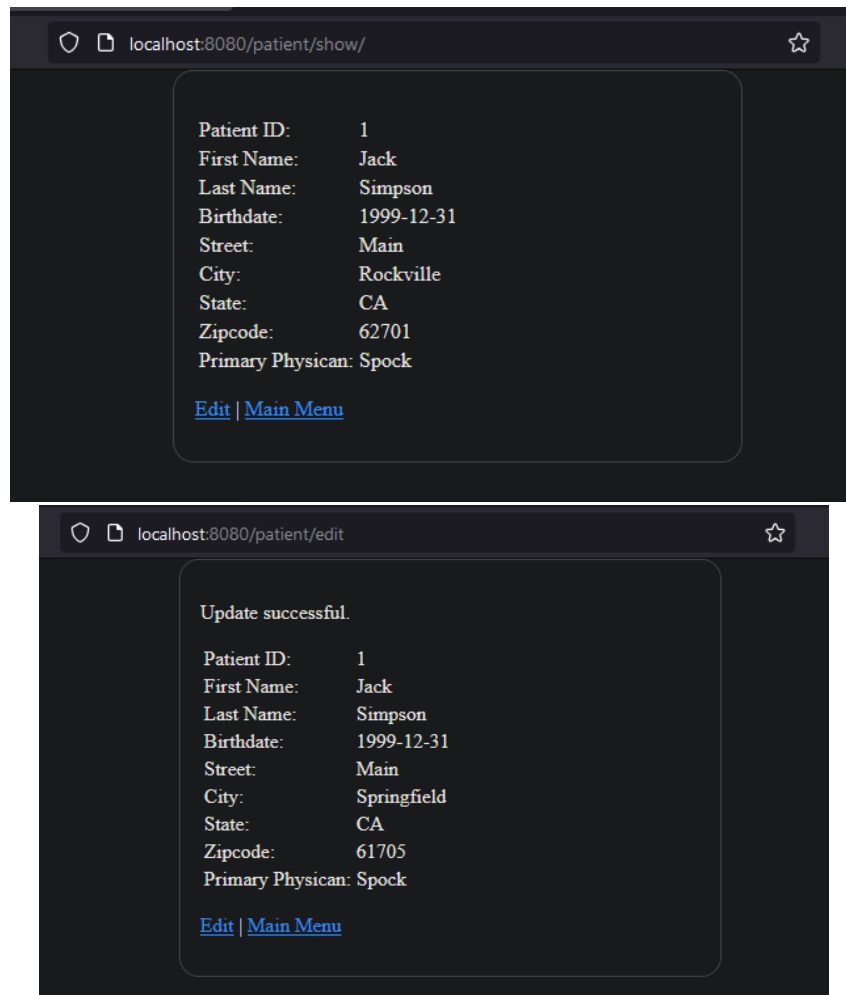
7. Successful prescription fill:

localhost:8080/prescription/fill

Rx: 2
Doctor ID: 0
First Name: Spock
Last Name: Spock
Patient ID: 1
First Name: Jack
Last Name: Simpson
Drug: Lisinopril
Quantity: 90
Refills remaining: 0
Pharmacy ID: 0
Name: CVS
Address: 111 Main Street
Phone:
Date Filled: 2024-10-08
Cost: \$

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8. Get the profile for patient “Simpson” and edit the patient record for “Simpson” and change city to Springfield and zip code to 61705. Show the web page of the successful update.



9. Edit the patient record for “Simpson”. Attempt to change the doctor’s name to a doctor that does not exist. Show the error message and edit patient form.

Update Patient Profile

Doctor not found.

ID: 1

First Name: Jack

Last Name: Simpson

BirthDate: 1999-12-31

Street: Main

City: Springfield

State: CA

Zipcode: 61705

Primary Physician
Name: Strange

Submit Change